The domain of the Project: Reinforcement Learning.

G35 PYTHON.

Team Mentors (and their designation):

Mr. Gaurav Patel

Team Members:
Ms. Rachi Rajpal

Period of the project

March 2026 to April 2026

Declaration

The project titled "Traffic Management System using Reinforcement Learning" has been mentored by Gaurav Patel, organized by SURE Trust, from January 2026 to March 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge, the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Ms. Rachi Rajpal ,

Mr. Gaurav Patel
Tutor, Sure Trust.

Prof. Radhakumari
Executive Director & Founder
SURE Trust

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Executive Summary

The **Traffic Management System using Reinforcement Learning** is an intelligent traffic signal control framework designed to optimize traffic flow at road intersections using Artificial Intelligence. Rapid urbanization and increasing vehicle density have made traditional fixed-time traffic signals inefficient, leading to congestion, long waiting times, and increased fuel consumption.

This project introduces a data-driven traffic control system that learns optimal signal timings using **Reinforcement Learning (RL)**. The system utilizes a simulation environment where a reinforcement learning agent observes traffic conditions such as vehicle queue length, waiting time, and lane occupancy. Based on these observations, the agent decides when to change traffic lights in order to minimize congestion and maximize vehicle throughput.

The system was implemented using **Python**, the **SUMO (Simulation of Urban Mobility) traffic simulator**, and **Stable-Baselines3**, a reinforcement learning framework. A **Proximal Policy Optimization (PPO)** algorithm was used to train the RL agent, enabling it to continuously improve its traffic signal control strategy through reward feedback.

To evaluate system performance, key traffic metrics such as **average waiting time, queue length, vehicle throughput, and cumulative reward trends** were analysed. The project also includes an **interactive Streamlit dashboard** to visualize real-time simulation metrics, training performance, and traffic analytics.

Experimental results demonstrate that reinforcement learning-based traffic control can significantly improve traffic efficiency compared to traditional fixed-time signal systems. The proposed system highlights the potential of AI-driven traffic management solutions for smart cities and intelligent transportation systems.

Introduction

Background and Context of the Project

Traffic congestion is a major problem in urban areas due to the rapid growth in vehicle population and limited road infrastructure. Traditional traffic signals operate on fixed timing schedules that do not adapt to real-time traffic conditions. As a result, vehicles often experience unnecessary delays and inefficient traffic flow.

Recent advancements in Artificial Intelligence and machine learning have opened new possibilities for building intelligent transportation systems. Reinforcement Learning (RL) is particularly suitable for traffic signal control because it allows systems to learn optimal decisions through interaction with the environment.

This project explores the application of reinforcement learning to dynamically control traffic signals and reduce congestion at road intersections.

Problem Statement / Goals of the Project

Conventional traffic signal systems are unable to adapt dynamically to changing traffic conditions. During peak hours, this results in long queues and inefficient traffic movement.

The primary goals of this project are:

- To design an **AI-based traffic signal control system** using reinforcement learning.
- To simulate realistic traffic conditions using the **SUMO traffic simulator**.
- To minimize **vehicle waiting time and queue length** at intersections.
- To improve **traffic flow and vehicle throughput** using intelligent signal timing.
- To visualize traffic performance using an **interactive monitoring dashboard**.

Scope and Limitations of the Project

The project focuses on **simulation-based traffic management** rather than real-world deployment. The system is designed for a **single-junction intersection environment**, where the reinforcement learning agent controls traffic lights based on traffic conditions.

Although the system demonstrates promising results in simulation, real-world deployment would require integration with physical traffic infrastructure, sensors, and city traffic control systems.

Innovation Component

The **Traffic Management System using Reinforcement Learning** introduces an intelligent approach to controlling traffic signals by integrating Artificial Intelligence with traffic simulation technologies.

Unlike traditional traffic signal systems that rely on **fixed-time scheduling**, the proposed system dynamically adapts signal timings based on real-time traffic conditions observed in the simulation environment. This allows the system to respond intelligently to varying traffic densities and congestion levels.

The key innovation of this project lies in the **application of Reinforcement Learning (RL)** for traffic signal optimization. A reinforcement learning agent continuously learns optimal traffic control strategies by interacting with the environment and receiving reward feedback based on traffic performance metrics such as queue length and vehicle waiting time.

Another innovative aspect is the **integration of the SUMO traffic simulator with machine learning frameworks**, enabling realistic traffic modelling combined with intelligent decision-making. This combination allows the system to simulate real-world traffic scenarios while training an AI model to manage traffic efficiently.

Additionally, the project includes an **interactive visualization dashboard developed using Streamlit**, which provides real-time insights into traffic metrics such as vehicle queue lengths, reward trends, and traffic flow performance. This enhances transparency and helps analyse the effectiveness of the reinforcement learning model.

Overall, the project demonstrates how **AI-driven adaptive traffic control systems** can outperform traditional traffic management approaches by continuously learning and improving from traffic patterns.

Project Objectives

Clearly Defined Objectives and Goals of the Project

The primary objective of the **Traffic Management System using Reinforcement Learning** project is to develop an intelligent traffic signal control mechanism that can dynamically adapt to changing traffic conditions and improve the efficiency of road intersections.
The key objectives of the project include:

- To design and implement an **AI-based traffic signal control system** using Reinforcement Learning techniques.
- To simulate realistic traffic scenarios using the **SUMO (Simulation of Urban Mobility)** traffic simulator.
- To train a reinforcement learning agent that can **learn optimal traffic signal timings through interaction with the simulation environment**.
- To minimize **vehicle waiting time and traffic congestion** at road intersections.
- To optimize **traffic flow and improve vehicle throughput** by dynamically adjusting signal phases.
- To analyse traffic performance metrics such as **queue length, waiting time, and reward trends** during training.
- To build a **visualization dashboard** that displays simulation results and training metrics in an interactive manner.
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Expected Outcomes and Deliverables

The expected outcomes of this project include the development of a functional prototype of an intelligent traffic management system capable of optimizing signal timings using reinforcement learning.
The key deliverables of the project are:

- A **simulation-based traffic management system** developed using reinforcement learning algorithms.
- A trained reinforcement learning model capable of making intelligent traffic signal decisions.
- Integration of the system with the **SUMO traffic simulator** to simulate real-world traffic conditions.
- A **Streamlit-based dashboard** to visualize traffic metrics such as queue length, vehicle waiting time, and training rewards.
- Graphical analysis and performance evaluation of the reinforcement learning model.
- Documentation of the system architecture, implementation methodology, and experimental results.

The project aims to demonstrate how **Artificial Intelligence can be applied to traffic signal control systems to improve transportation efficiency and reduce urban congestion**.

Methods / Technology Used

The project uses **Reinforcement Learning (RL)** to develop an intelligent traffic signal control system. A reinforcement learning agent interacts with a simulated traffic environment and learns optimal signal control strategies based on reward feedback.

The **Proximal Policy Optimization (PPO)** algorithm was used for training because it provides stable policy updates and efficient learning in complex environments.

The agent observes several traffic parameters including:

- Vehicle queue length
- Waiting time of vehicles
- Lane density
- Traffic signal state

Based on these observations, the agent decides whether to **maintain the current traffic signal phase or switch to another phase**.

The reward function was designed to encourage efficient traffic flow by minimizing congestion and reducing vehicle waiting times.

Tools / Software Used

Languages:

- Python

Frameworks / Libraries:

- Stable-Baselines3 (Reinforcement Learning)
- SUMO Traffic Simulator
- Gymnasium Environment
- NumPy
- Pandas
- Matplotlib
- Streamlit

Development Tools:

- Git & GitHub for version control
- Visual Studio Code for development

Project Repository:

Data Collection Approach

Traffic data for the project was generated using the **SUMO simulation environment**. SUMO allows realistic modelling of vehicle movements, road networks, and traffic signals.

Simulation scenarios were created with varying traffic densities to evaluate system performance under different traffic conditions. Data such as vehicle waiting time, queue length, and traffic flow rates were collected during simulation runs.

Project Architecture

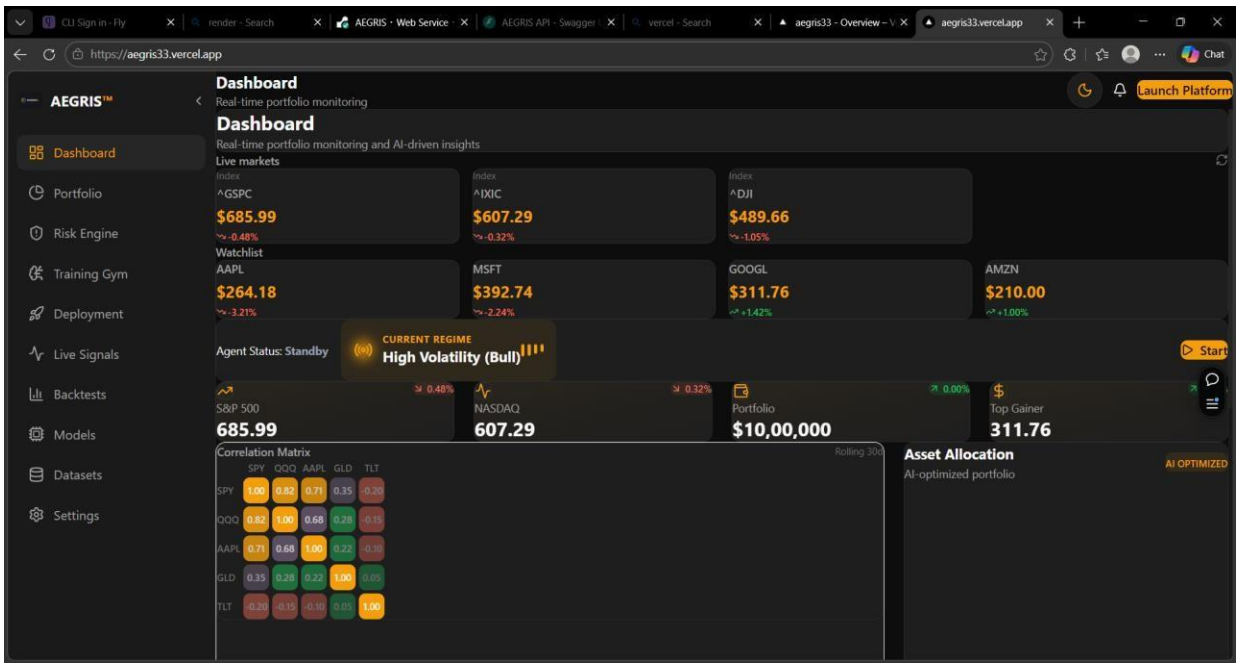
The system follows a modular architecture consisting of the following components:

1. **Traffic Simulation Layer**
   SUMO simulates the road network, vehicle movements, and traffic signal operations.
2. **Reinforcement Learning Agent**
   The RL agent observes traffic states and determines optimal signal control actions.
3. **Training Module**
   The PPO algorithm trains the RL agent using reward signals derived from traffic performance metrics.
4. **Visualization Dashboard**
   A Streamlit-based dashboard displays real-time traffic statistics, queue lengths, rewards, and simulation performance

Final Project Working

During simulation, vehicles approach an intersection from multiple directions. The reinforcement learning agent continuously monitors the traffic state and dynamically adjusts signal timings.

As the training progresses, the agent learns policies that reduce traffic congestion and improve traffic flow efficiency. The dashboard provides visual insights into the training progress and traffic performance metrics.



Learning and Reflection

The development of the Traffic Management System using Reinforcement Learning provided valuable experience in multiple areas of Artificial Intelligence and system design.

From a technical perspective, the project strengthened understanding of reinforcement learning algorithms, particularly policy-based methods such as Proximal Policy Optimization. It also provided hands-on experience with simulation environments such as SUMO and reinforcement learning frameworks like Stable-Baselines3.

Working on this project improved skills in data analysis, algorithm implementation, and system integration. Designing reward functions and training reinforcement learning agents required careful experimentation and evaluation.

Several challenges were encountered during development, including designing an effective reward structure, managing simulation performance, and ensuring stable training of the reinforcement learning model. Overcoming these challenges helped improve problem-solving and analytical thinking skills.

Overall, the project provided practical exposure to building AI-driven intelligent systems and highlighted the real-world potential of reinforcement learning in smart transportation systems.

Conclusion and Future Scope

Future Scope

The Traffic Management System can be further enhanced in several ways:

- Integration with **real-time traffic sensors and IoT devices**
- Deployment across **multiple interconnected intersections**
- Use of **multi-agent reinforcement learning** for city-scale traffic management
- Integration with **computer vision systems** for vehicle detection
- Development of **predictive traffic analytics using deep learning models**

These improvements would help transform the system into a fully functional smart city traffic management solution.

Conclusion

The Traffic Management System using Reinforcement Learning demonstrates the effectiveness of AI-driven approaches for intelligent traffic signal control. By dynamically adapting traffic signal timings based on real-time traffic conditions, the system can significantly reduce congestion and improve overall traffic efficiency.

The project successfully integrates reinforcement learning algorithms with traffic simulation environments and interactive dashboards, creating a scalable prototype for intelligent transportation systems.

The results highlight the potential of reinforcement learning as a powerful tool for solving complex real-world optimization problems such as urban traffic management.

Research Basis

The project is based on concepts from:

- Reinforcement Learning
- Intelligent Transportation Systems
- Traffic Flow Optimization
- Artificial Intelligence for Smart Cities

Technical References

1. Sutton, R. S., & Barto, A. G. – *Reinforcement Learning: An Introduction*
2. PyTorch Official Documentation – https://pytorch.org
3. Stable-Baselines3 Documentation – https://stable-baselines3.readthedocs.io
4. SUMO Traffic Simulator Documentation
5. Stable-Baselines3 Documentation
6. OpenAI Gym / Gymnasium Documentation
7. Research papers on Reinforcement Learning for Traffic Signal Control

Tools and Open-Source Libraries Referenced

The development of the **Traffic Management System using Reinforcement Learning** involved several programming tools, frameworks, and open-source libraries that supported simulation, machine learning model development, and data visualization.

Programming Language

- Python – Used for implementing reinforcement learning algorithms, simulation integration, and data processing.

Traffic Simulation

- SUMO (Simulation of Urban Mobility) – Used to simulate realistic traffic environments, vehicle movements, and traffic signal operations.

Machine Learning and Reinforcement Learning

- Stable-Baselines3 – Used to implement reinforcement learning algorithms such as Proximal Policy Optimization (PPO).
- Gymnasium / OpenAI Gym – Used for creating reinforcement learning environments and integrating them with training frameworks.

Data Processing and Analysis

- NumPy – Used for numerical computations and matrix operations.
- Pandas – Used for handling and analysing traffic simulation data.

Visualization and Monitoring

- Matplotlib – Used for generating graphs and visualizing training metrics.
- Streamlit – Used to build an interactive dashboard for monitoring traffic simulation results and reinforcement learning performance.

Development Tools

- Visual Studio Code – Used as the primary development environment.
- Git & GitHub – Used for version control and project management.

These tools collectively enabled the development of a complete system that integrates **traffic simulation, reinforcement learning, data analysis, and visualization** into a unified intelligent traffic management framework.

Project Repository

Traffic Management System using Reinforcement Learning GitHub Repository: